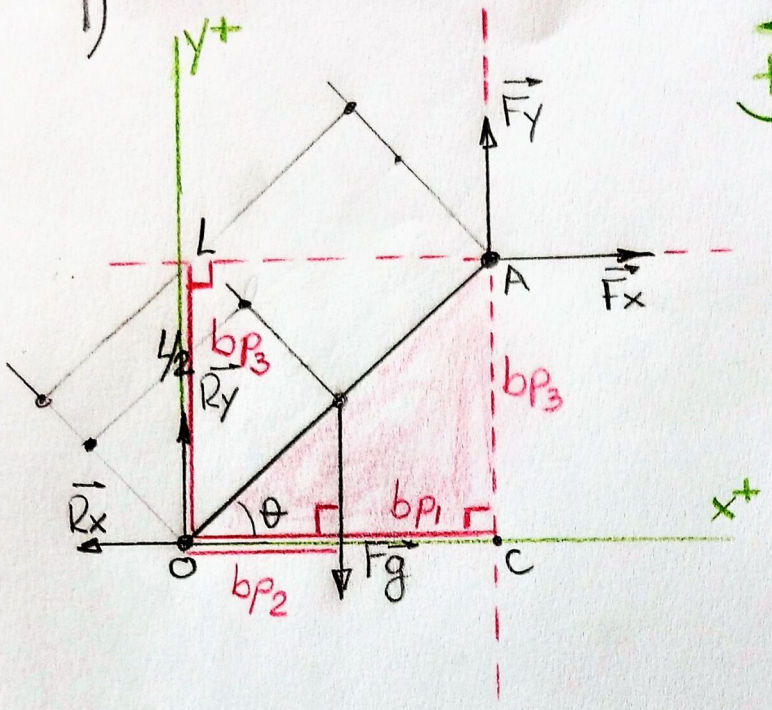
UNIDAD III Equilibrio Estático

1)

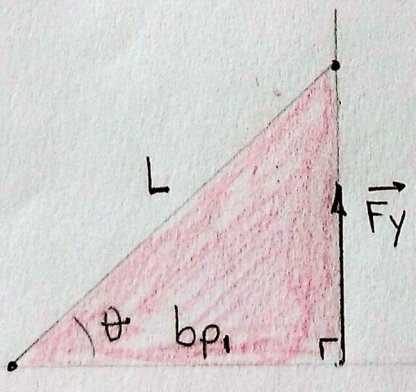


$$\sum F_x: \quad F_x - R_x = 0$$

$$\sum F_y: \quad F_y + R_y - F_g = 0$$

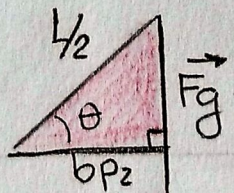
$$\sum \tau_O: \quad F_y . bp_1 - F_g\, bp_2 - F_x\, bp_3 = 0$$

$$F_y . L . \cos\theta - F_g \frac{L}{2}\cos\theta - F_x L \operatorname{sen}\theta = 0$$



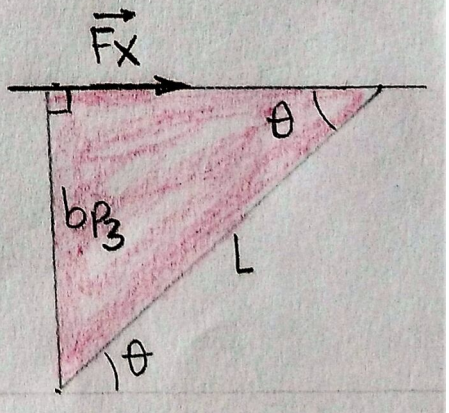
$$\cos\theta = \frac{bp_1}{L}$$

$$bp_1 = L . \cos\theta$$



$$\cos\theta = \frac{bp_2}{L/2}$$

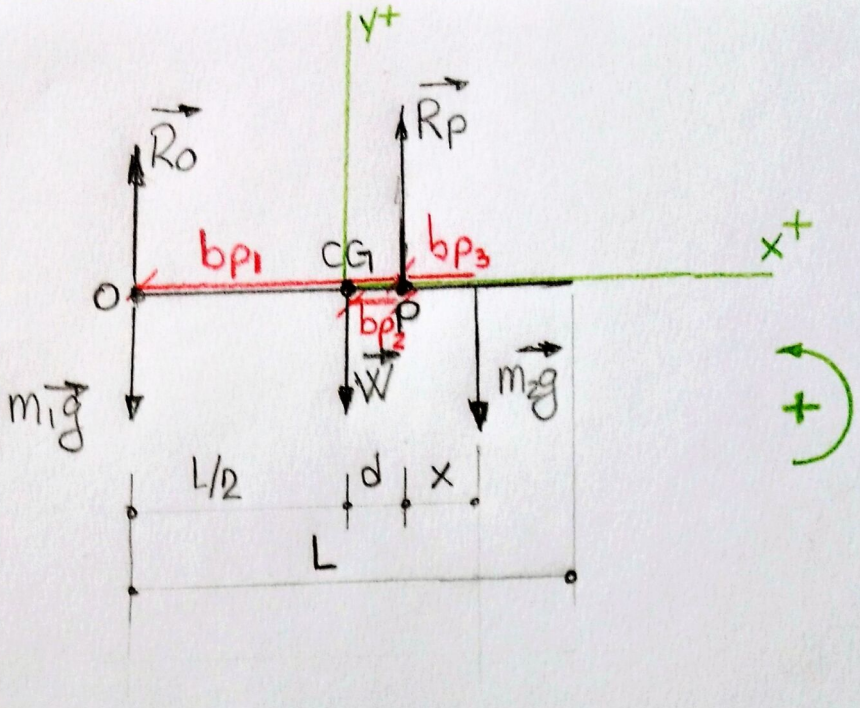
$$bp_2 = \frac{L}{2} . \cos\theta$$



$$\operatorname{sen}\theta = \frac{bp_3}{L}$$

$$bp_3 = L \operatorname{sen}\theta$$

2)



$$\sum F_x = 0$$

$$\sum F_y : \cancel{R_O^{\,0}} + R_P - m_1 g - W - m_2 g = 0$$

$$\sum \vec{\sigma}_P : \; m_1 g \cdot b_{P1} + W b_{P2} - m_2 g\, b_{P3} = 0$$

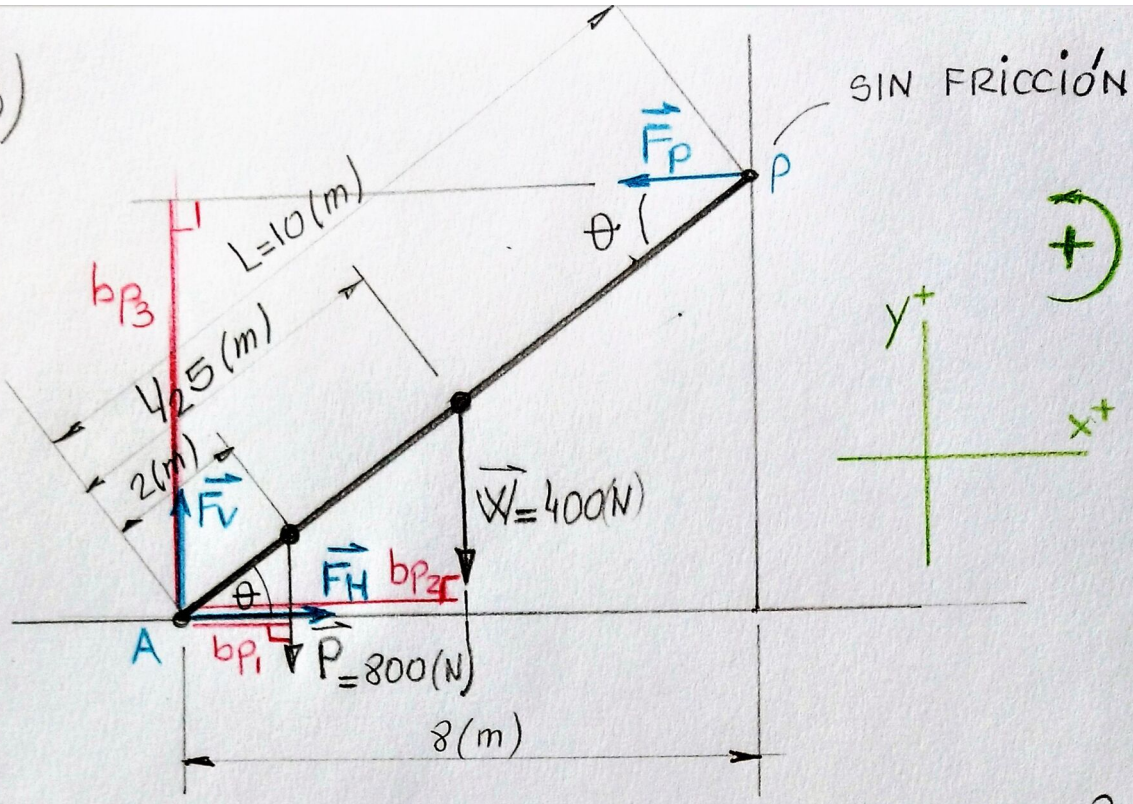
$$m_1 g\left(\frac{L}{2} + d\right) + W \cdot d - m_2 g \cdot x = 0$$

$$m_1 g \frac{L}{2} + W d = m_2 g \cdot x$$

$$x = \frac{m_1 g \frac{L}{2} + m_1 g \cdot d + W \cdot d}{m_2 g}$$

$$x = \frac{(m_1 g + W) d + m_1 g \frac{L}{2}}{m_2 g}$$

3)



$$\cos\theta = \frac{8}{10}$$

$$\theta = 36{,}9$$

$$\sum F_x: \quad F_H - F_P = 0$$

$$\sum F_y: \quad F_V - P - W = 0$$

$$\sum \mathcal{T}_A: \quad F_P \cdot b_{P_3} - P \cdot b_{P_1} - W \cdot b_{P_2} = 0$$

$$\operatorname{sen}\theta = \frac{b_{P_3}}{L} \qquad \cos\theta = \frac{b_{P_1}}{2} \qquad \cos\theta = \frac{b_{P_2}}{L/2}$$

$$F_P \, L \cdot \operatorname{sen}\theta - P \cdot 2 \cdot \cos\theta - W \cdot \frac{L}{2}\cos\theta = 0$$

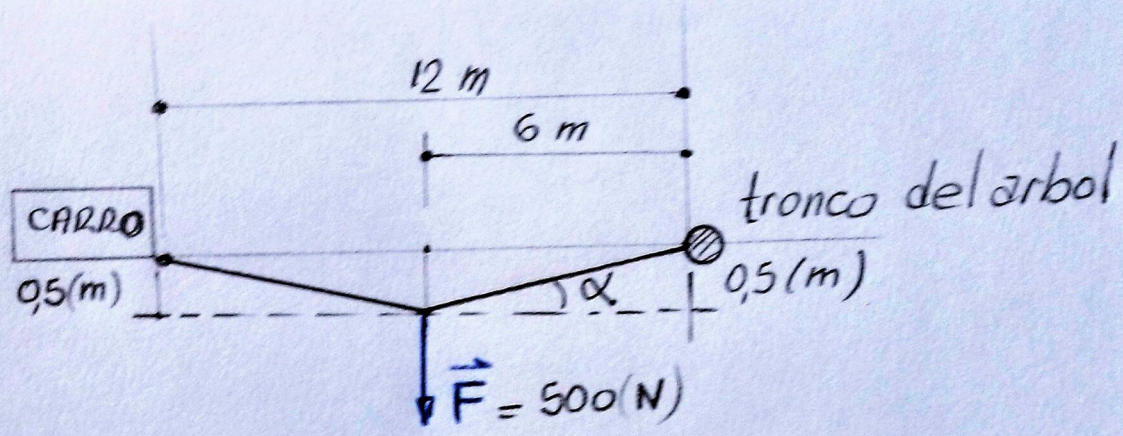
$$F_P = \frac{\cos\theta\left(2P + W \cdot \frac{L}{2}\right)}{L \operatorname{sen}\theta} =$$

$$F_P = \frac{\cos 36{,}9 \cdot \left(2 \cdot 800 + 400 \cdot \frac{10}{2}\right)}{10 \cdot \operatorname{sen} 36{,}9} = 479{,}5\,(N)$$

$$F_H = F_P = 479{,}5\,(N).$$

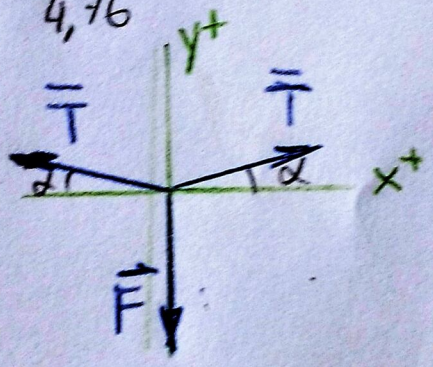
$$F_V = P + W = 800 + 400 = 1200\,(N)$$

4)



$$tg\,\alpha = \frac{0,5}{6} = 0,08\hat{3}$$

$$\alpha = 4,76°$$



$$\sum Fx: \quad T\cos\alpha - T\cos\alpha = 0$$

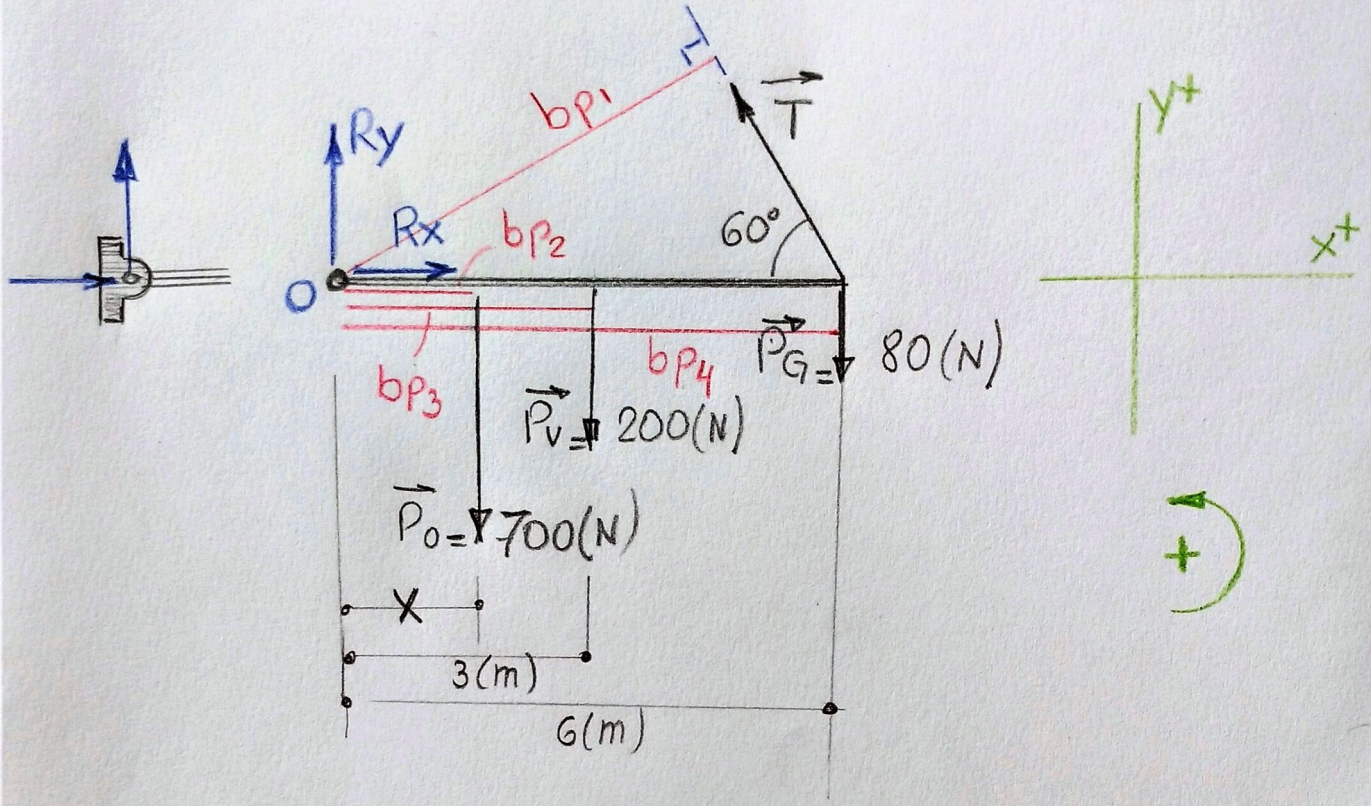
$$\sum Fy: \quad 2T\,sen\alpha - F = 0$$

$$F = 2T\,sen\alpha$$

$$T = \frac{F}{2\,sen\alpha} = \frac{500}{2.sen\,4,76}$$

$$T = 3010,4\,(N)$$

9)



$$\sum Fx: \quad Rx - T\cos 60 = 0$$

$$\sum Fy: \quad Ry + T\,sen\,60 - P_0 - P_V - P_G = 0$$

$$\sum \bar{C}_0: \quad T.b_{P_1} - P_0.b_{P_2} - P_V.b_{P_3} - P_G.b_{P_4} = 0$$

$$T.6.sen\,60° - P_0 X - P_V.3 - P_G.6 = 0$$

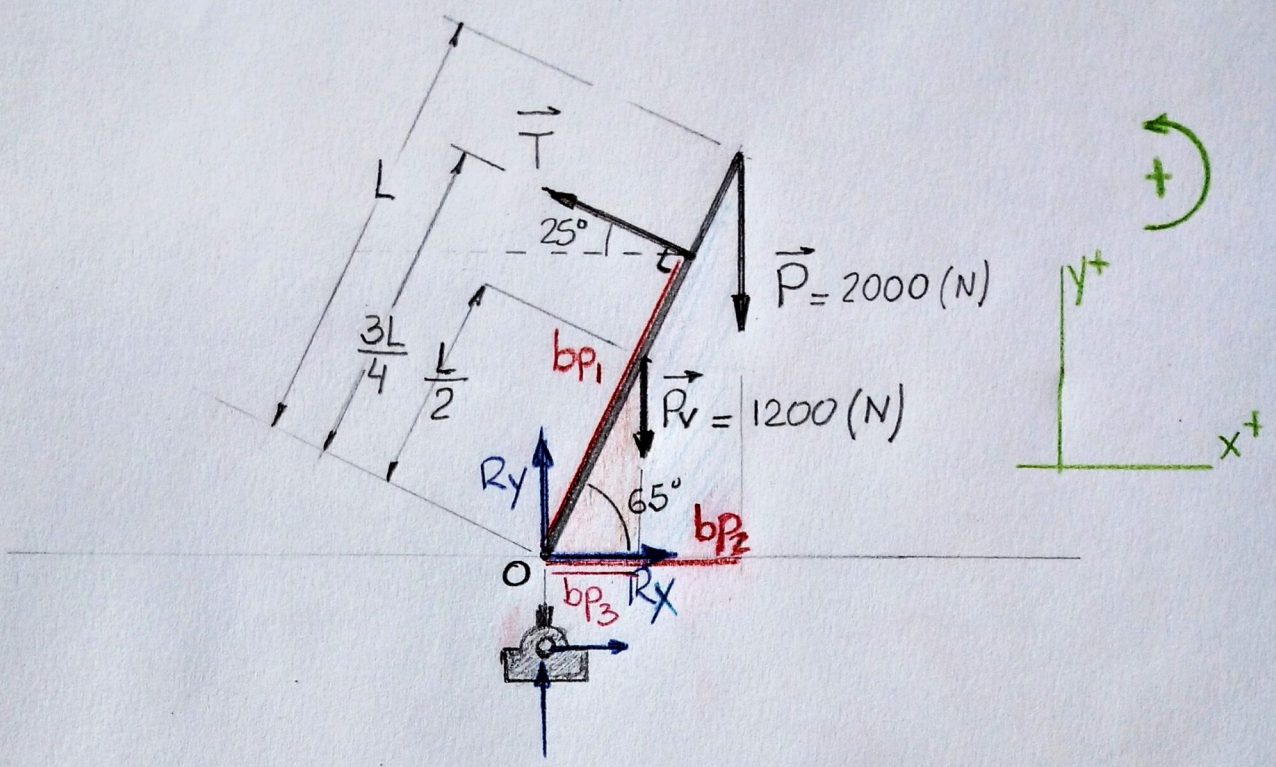
$$T\Big|_{X=1(m)} = \frac{700.1 + 200.3 + 80.6}{6\,sen\,60} = 342{,}6\,(N)$$

$$Rx = T\cos 60 = 342{,}6 \cdot \cos 60 = 171{,}3\,(N)$$

$$Ry = P_0 + P_V + P_G - T\,sen\,60 = 700 + 200 + 80 - 342{,}6\,sen\,60 =$$

$$Ry = 683{,}3\,(N)$$

10)



$$\sum Fx: \quad Rx - T\cos 25 = 0$$

$$\sum Fy: \quad Ry + T\,sen\,25 - Pv - P = 0$$

$$\sum \mathcal{Z}_o: \quad T.bp_1 - P.bp_2 - Pv.bp_3 = 0$$

$$\cos 65° = \frac{bp_3}{\frac{L}{2}} \qquad \cos 65° = \frac{bp_2}{L}$$

$$\sum \mathcal{Z}_o: \quad T.\frac{3L}{4} - P.L\cos 65° - Pv.\frac{L}{2}\cos 65° = 0$$

$$T = \frac{P\cos 65° + Pv\frac{1}{2}\cos 65}{\frac{3}{4}} =$$

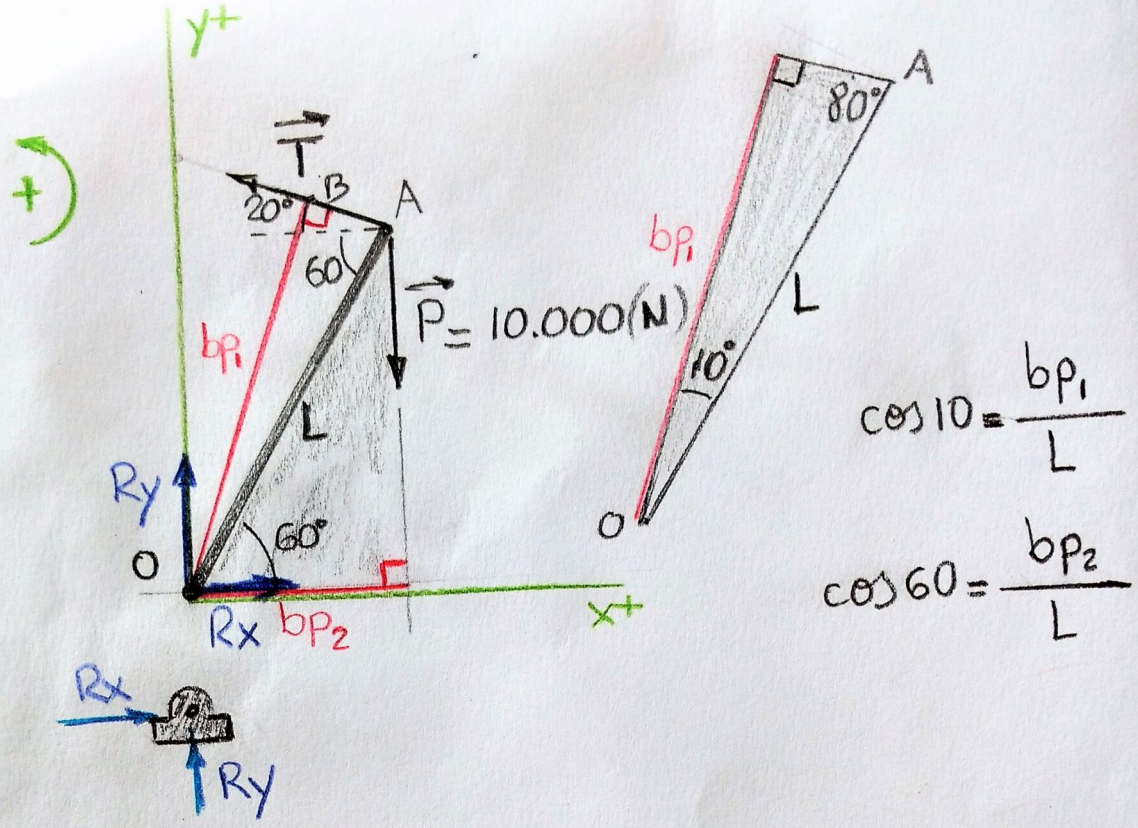
$$T = \frac{4.\cos 65°\,(2000 + \frac{1200}{2})}{3} = 1465,1\,(N)$$

$$Rx = T\cos 25 = 1465,1.\cos 25 = 1327,8\,(N)$$

$$Ry = Pv + P - T\,sen\,25 = 1200 + 2000 - 1465,1\,sen\,25°$$

$$Ry = 2580,9\,(N)$$

12)



$\Sigma Fx: \quad Rx - T\cos 20 = 0$

$\Sigma Fy: \quad Ry + T\,sen\,20 - P = 0$

$\Sigma \mathcal{T}_o: \quad T.bp_1 - P.bp_2 = 0$

$T.L.\cos 10^\circ - P.L.\cos 60^\circ = 0$

$$T = \frac{P.\cancel{L}.\cos 60}{\cancel{L}.\cos 10} = \frac{10000\cos 60}{\cos 10} =$$

$T = 5077,1\ (N)$

$Rx = T\cos 20 = 5077,1.\cos 20 = 4770,9(N)$

$Ry = P - T\,sen\,20 = 10.000 - 5077,1\,sen\,20$

$Ry = 8263,5\ (N).$